

Day 3: Preprocessing concepts

Lecturer: Msc. Minh Tan Le

Plans

- I. Simulation with distribution
- II. Data description & conclusion
- III. Preprocessing
- IV. Use case: Why is my bus always late?

I. Simulation with distribution

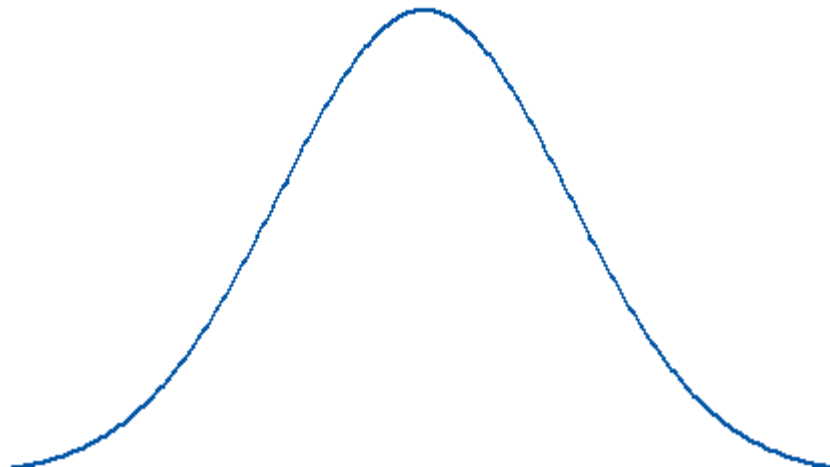
choice/choices

Distributions

- Normal/Gauss
- Bernulli
- Binomial
- Hypergeometric

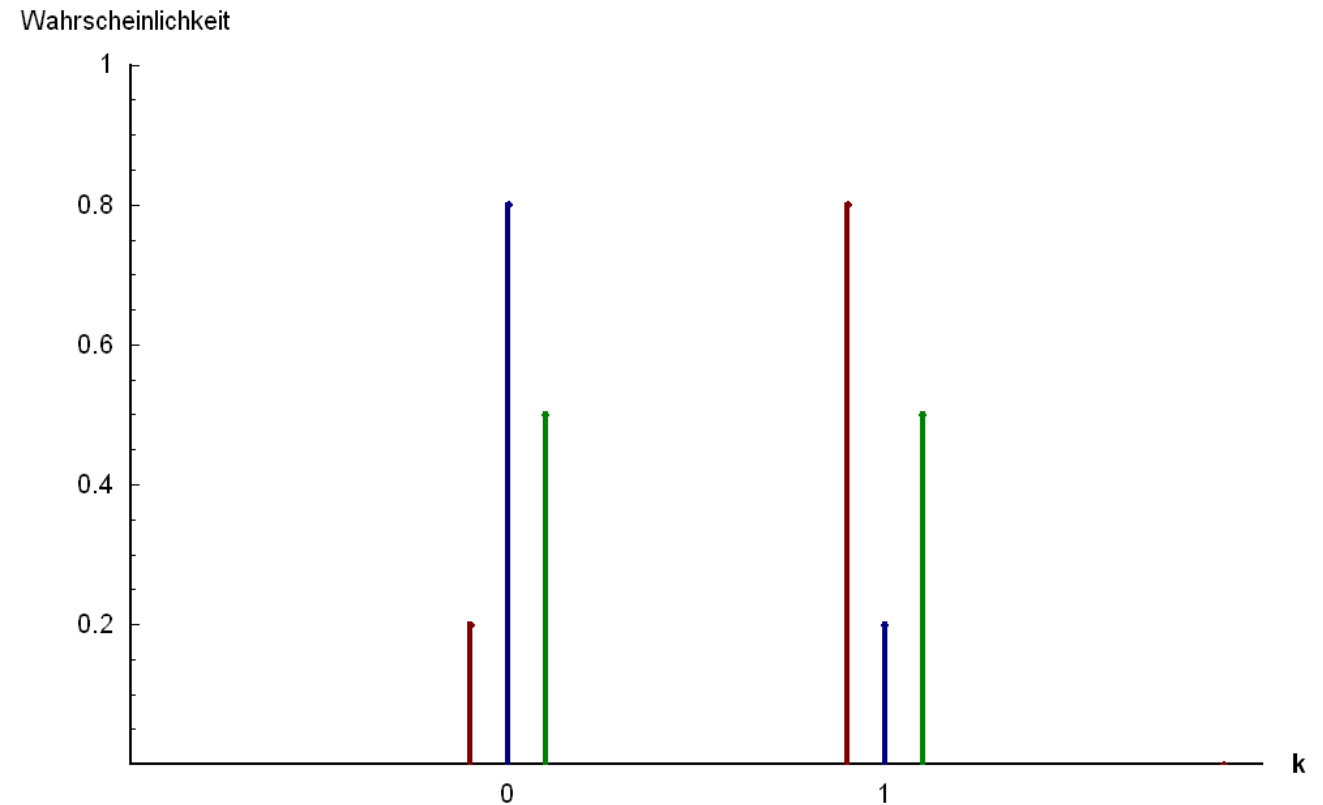
Normal/Gauss

- Used for continuous, multiple data.
- Popular in natural & social sciences when the distributions are unknown.
- *Examples: Height data of first-grade students.*



Bernulli

- Yes/no question.
- *Example: Coin flipping*



■ $P(x = 0) = 0.2$ and $P(x = 1) = 0.8$
■ $P(x = 0) = 0.8$ and $P(x = 1) = 0.2$
■ $P(x = 0) = 0.5$ and $P(x = 1) = 0.5$

Binomial

- Getting k successes in n independent trials.
- Sampling with replacement.

$$pmf(k, n, p) = C_k^n p^k (1 - p)^{n-k}$$

Hypergeometric

- Getting k successes in n dependent trials.
- Work with non-replacement sampling (simple random sample).
- Dependent sampling.

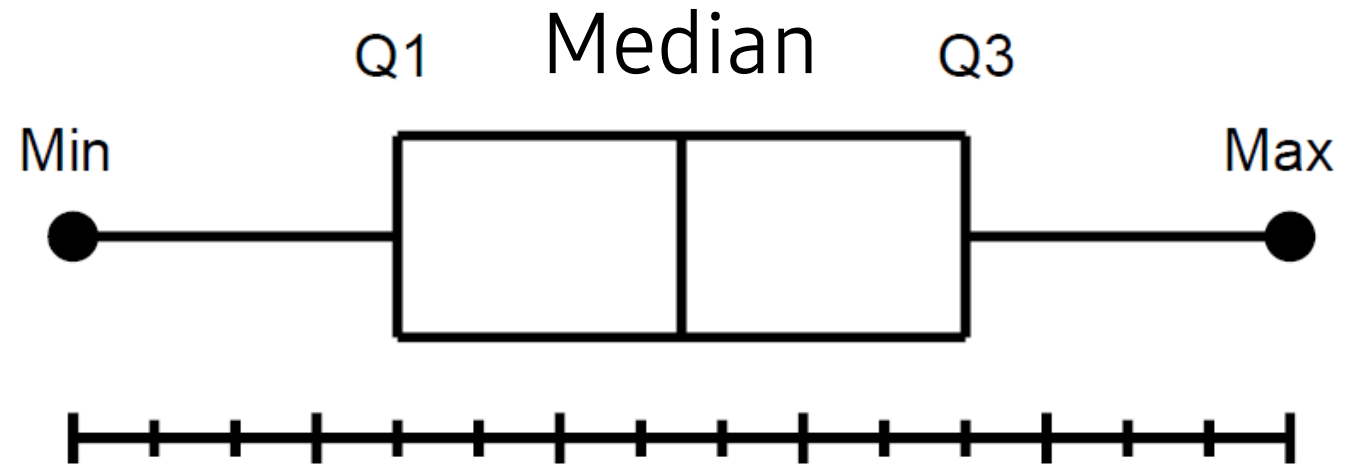
Demo

II. Data description

- Describing the data should includes:
 - Common statistics
 - Moment statistics
 - Scoped statistics

Common statistics

- Quantity (count/size)
- Max (maximum/maxima)
- Min (minimum/minima)
- Mean (average)
- Standard deviation (std)
- Median
- Quartile (percentile)



Moment statistics (of a function)

- Mean
- Variance
- Skewness
- Kurtosis
- Hyperskewness
- Hypertailedness

Scoped statistics

- You are building an insurance product recommendation system.
- The suggestion should be based on client data only.

Client	Agent	Product
Le Minh Tan	Nguyen Thi A	BH Nhan Tho A1
Le Minh Tan	Le Thi B	BH Tai Nan A2
...	...	...

- You found out: Each agent is bias of recommending a product.

Modeling with statistics

- Define **null hypothesis** (which is assumptions).
- Build related model/functions (mostly parametric).
- Conclude **alternative hypothesis**.

Abnormal statistics to model



- Mean absolute error (MAE)
- Mean square error (MSE)

III. Preprocessing

- Data cleaning
- Data integration
- Data transformation
 - Data scaling
 - Data Discretization
- Feature engineering

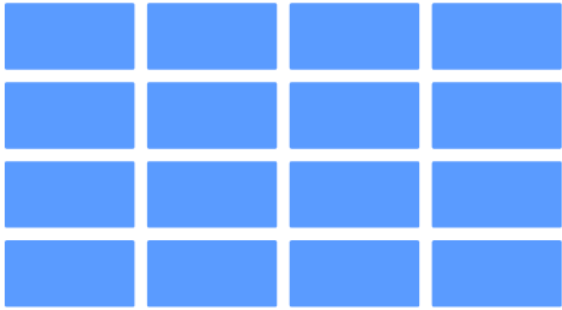
Cleaning tasks

1. Remove duplicate & irrelevant
2. Improve data structural consistency
3. Remove outliers
4. Missing data processing
5. Last one: Discussing

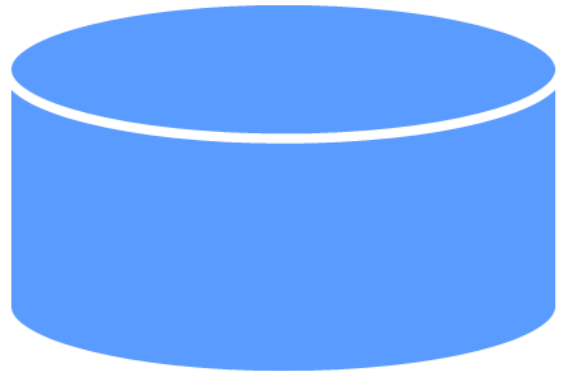
Data integration

- Purpose: Combine multiple sources into a unified format.
- Solution:
 - ETL (extract, transform, load).
 - ELT (extract, load, transform).

Extract



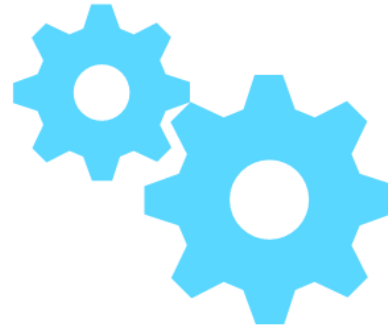
Data source 1



Data source 2



Transform



Transformation
engine

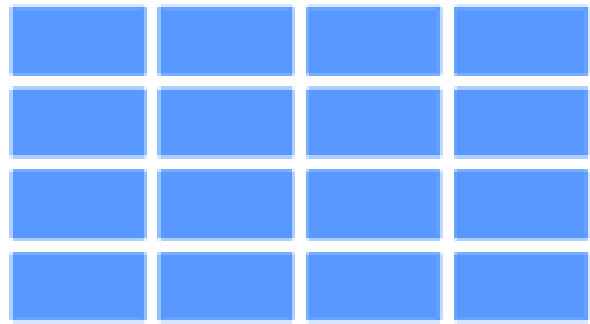


Load

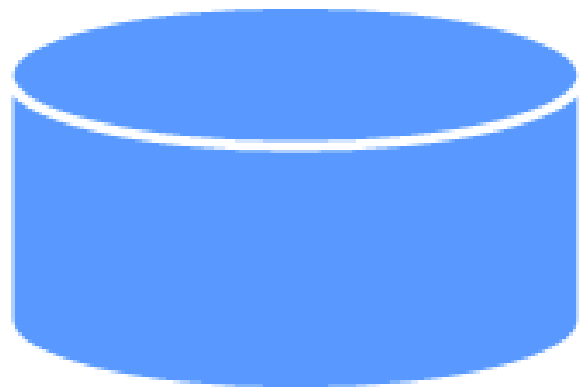


Target

Extract



Data source 1



Data source 2

Load

Transform



Target



Data transformation

- Formatting
 - 17/03/2003 -> 2003-03-17
- Filtering
 - Target: gen-Z youth -> 1997 – 2012
- Aggregation
 - Sum, avg, min, max,... with group by

Data/feature scaling

- Goal: Min-max of a field stays in specific ranges.
- Common types:
 - Standardization: $\mu = 0, \sigma = 1$
 - Normalization: $0 \leq x \leq 1$
 - Min-max scaling: $\min = 0, \max = 1$

Data discretization

- Purpose: Data is separated into groups.
- Techniques
 - Histogram analysis
 - Cluster analysis
 - Correlation analysis

The bus time



Buses to School

10:00 10:04 10:08 10:20 10:24
10:36 10:40 10:44 10:48
11:00 ...

(9 buses come each hour)

⋮

9 buses come each hour

- If I go to the bus stop at random time, I'm expected to wait for ~6.7 minutes to catch a bus.
- Is it correct?

Case study: Waiting time paradox

If an event occurs at a time that we pick uniformly at random, the average waiting time until the next event is typically larger than half of the inter-event time.

A

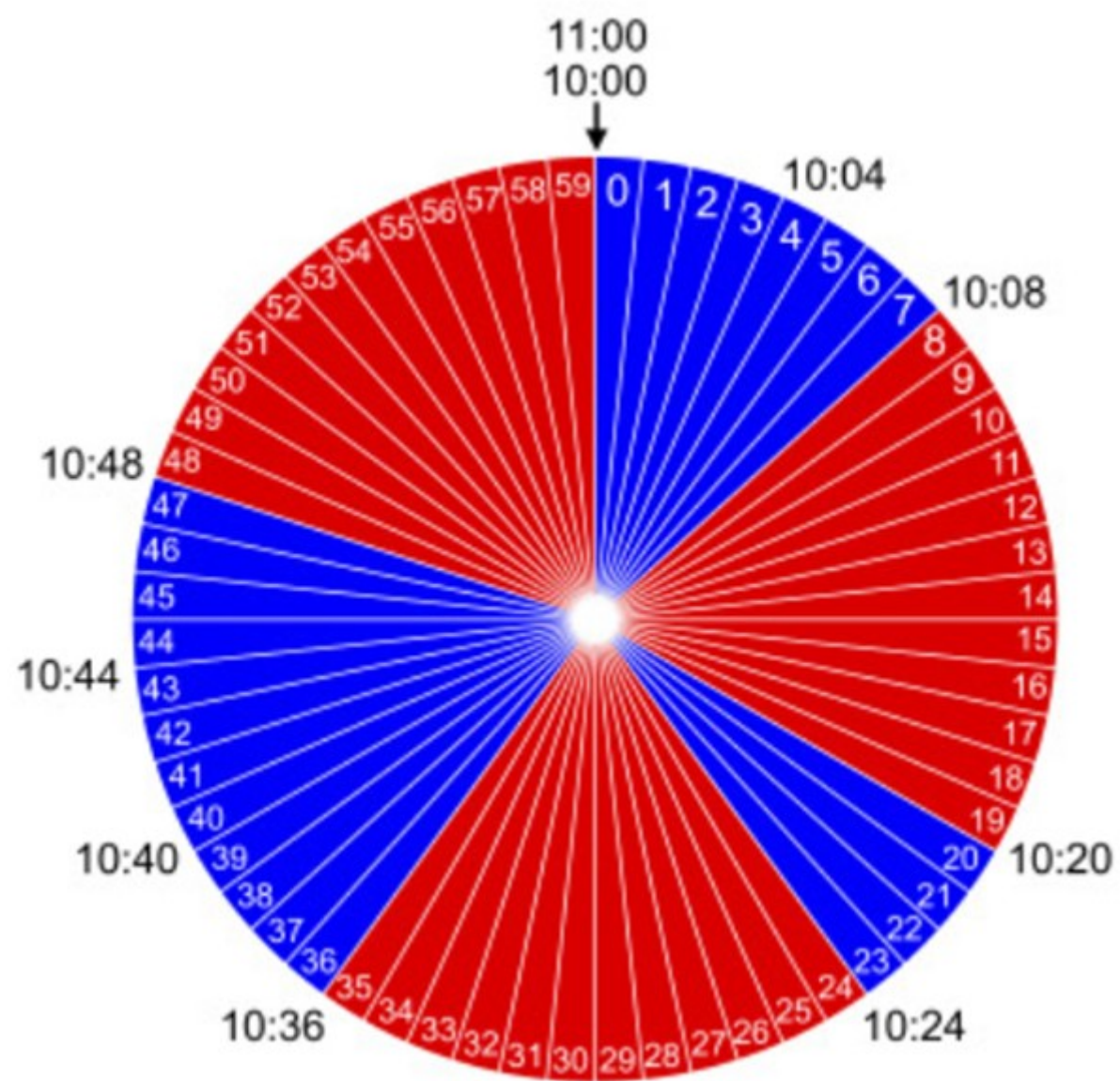
Buses to School			
Every hour	:00	:04	:08
	:20	:24	:36
	:40	:44	:48

I think I'll get a 4-minute inter-event time because it occurs more often.



B

Inter-event time	Number of times	Probability
↔ 4 minutes	6	$\frac{6}{6+3} = \frac{2}{3}$, so $\approx 67\%$
↔ 12 minutes	3	$\frac{3}{6+3} = \frac{1}{3}$, so $\approx 33\%$



Inter-event time	Probability
↔ 4 minutes	$\frac{24}{24 + 36} = \frac{2}{5}$, so 40%
↔ 12 minutes	$\frac{36}{24 + 36} = \frac{3}{5}$, so 60%



Wow! I was wrong. In fact, I am more likely to get ↔ 12 minutes, even though 12-minute inter-event times occur less often.

<https://jakevdp.github.io/blog/2018/09/13/waiting-time-paradox/>